How to use: smFISH_compartments_and_areas.py (and smFISH_dendrite_vs_axon.py)

Dendrite vs axon spot density with both analyses, and a check of compartment areas / culture confluence. pipeline version 4 (October 2026).

Input: counts_per_image.csv. **Needs:** smFISH_statistics_superplots.py in the same folder (shares its statistics settings and plot style).

1. Running it

```
python smFISH_compartments_and_areas.py
-> select counts_per_image.csv
```

Results: `_density_results/compartments_and_areas/`.

2. Part 1 - dendrite vs axon

- **t-test:** paired (dendrite and axon from the same images) on log10 replicate means, per probe × timepoint × condition.
- **Robust model:** per timepoint × condition one model across all probes, mean ~ probe × compartment + differentiation; dendrite-vs-axon contrast per probe.
- Output: SuperPlots in t_tests/ and robust_linear_model/; first Excel sheet = both methods side by side.
- Note: the "axon" mask contains all non-dendritic neurites in the field, including axons of neighbouring cells.

`smFISH_dendrite_vs_axon.py` is a standalone version of Part 1 only. It writes to `statistics_and_plots/dendrite_vs_axon/` and uses the same tests. Use either; the results are identical.

3. Part 2 - areas and confluence

Output	Question it answers
AREA_per_sample_<compartment>.pdf	Is any sample (probe × replicate × condition) more or less confluent? Red "!" = median area outside 1.5× of the timepoint median
sheet area_which_factor	Do areas differ by differentiation (expected), probe or condition (warning sign)?
CONFLUENCE_vs_density.pdf, sheet confluence_vs_density	Does confluence change the density? Elasticity 0 = no influence, 1 = density proportional to axon area. Estimated within control/protease pairs, so differentiation differences cannot create a false association

4. Settings

Setting	Default	Meaning
COMPARE	('Dendrite','Axon')	ratio reported as first / second
CONDITIONS	['control','protease']	conditions tested
CONFLUENCE_COMPARTMENT	'Axon'	compartment whose area measures confluence
AREA_FLAG_FOLD	1.5	flag threshold for per-sample areas